

Part A

Question 1:

$$F : C[0, 1] \rightarrow C[0, 1]$$

$$F(f)(t) = \int_t^1 (1 - x^4) f(x) \, dx \text{ for each } f \in C[0, 1] \text{ and } t \in [0, 1]$$

Let $f \in C[0, 1]$ and $s, t \in [0, 1]$ with $s \leq t$.

Let $f, g \in C[0, 1]$ and $t \in [0, 1]$.

$$\begin{aligned} |F(f)(t) - F(g)(t)| &= \left| \int_t^1 (1 - x^4) f(x) \, dx - \int_t^1 (1 - x^4) g(x) \, dx \right| \\ &= \left| \int_t^1 (1 - x^4) (f(x) - g(x)) \, dx \right| \\ &\leq \int_t^1 |(1 - x^4)(f(x) - g(x))| \, dx \\ &\leq \int_t^1 (1 - x^4) |f(x) - g(x)| \, dx \end{aligned}$$

Since $0 < (1 - x^4) < 1$ for $x \in [0, 1]$

$$\begin{aligned} &< \int_t^1 |f(x) - g(x)| \, dx \\ &< \int_t^1 d_{\max}(f, g) \, dx \\ &= d_{\max}(f, g)(1 - t) \\ &\leq d_{\max}(f, g)(1 - 0) \\ &= d_{\max}(f, g) \end{aligned}$$

Hence F is a $d_{\max}, d_{\max}$ -Lipschitz function on $C[0, 1]$. Thus F is continuous by Theorem 3.2 Chapter 15 p76.

